

AI Personal Assistant Complete User Guide

Visual Feature Walkthrough with Real Screen Previews, Multi-Provider API Setup Manual (OpenRouter, Gemini, Groq, DeepSeek), Model Hub Navigation, and Autonomous Scheduling.

✦ CORE SYSTEM CAPABILITIES

- Hermes Multi-Provider API Architecture
- 450+ Cloud Models Directory & Instant Search
- Free Tier LLM Hub (Nemotron, Ling, Qwen)
- On-Device Deterministic NLP Action Parser
- Autonomous AlarmManager Exact Reminders
- Local Room Database (Tasks, Notes, Chats)

PACKAGE / APPLICATION ID
com.mitaoe.shridhar202401040197

TARGET OS COMPATIBILITY
Android 8.0 to Android 14+

APPLICATION VERSION
1.0 (Release / Debug)

UI DESIGN LANGUAGE
Obsidian Cybernetic (Dark Neon)

Table of Contents

Chapter	Content & Feature Walkthrough	Page Reference
Chapter 1	How to Add API Keys & Where to Find Free / Paid APIs (OpenRouter, Gemini, Groq, etc.)	Page 3
Chapter 2	Model Library & Search Hub (Browsing 450+ Models, Free Tier, Pinning Models)	Page 4
Chapter 3	AI Chat Command Center & Quick Model Switcher (In-Chat Switching, Streaming, Copy, TTS)	Page 5
Chapter 4	Cyber Dashboard & Audio Synthesis (Greeting, Productivity Stats, Audio Briefing)	Page 6
Chapter 5	AI Workspace & Navigation Drawer (Profile, Conversation History, + New Chat)	Page 7
Chapter 6	Task Matrix & Autonomous Alarm Scheduling (Priorities, Reminders, Alarms)	Page 8
Chapter 7	Knowledge Vault (Notes) & AI Extraction Tools (Summarization, Extract Tasks to Matrix)	Page 9
Chapter 8	Natural Language Voice Commands & System Troubleshooting (Cheat Sheet & Permissions)	Page 10

System Architecture: The Dual-Engine Advantage

The AI Personal Assistant is uniquely engineered to provide uninterrupted utility regardless of your network connectivity or API quota status:

1. Local Deterministic Engine (Offline)

Powered by `NaturalLanguageActionParser.java`. Runs 100% locally on your phone with **zero API key and zero internet required**. It automatically intercepts commands like *"Remind me to call professor tomorrow at 3pm"*, saves them to Room SQLite, and sets exact hardware alarms.

2. Cloud Generative LLM Engine (Online)

Powered by `AIService.java` with Hermes-style multi-provider support. Allows you to plug in API keys from **OpenRouter, Google Gemini, Groq, DeepSeek, or OpenAI** to gain access to 450+ conversational reasoning models with live token streaming.

 **Privacy by Design:** The application stores all notes, tasks, alarms, and chat histories inside an encrypted local Room SQLite database on your device. Your personal data is never sent to third-party tracking or analytics servers.

1. How to Add API Keys & Where to Find Them

To unlock online cloud conversational intelligence, you can connect one or more API providers. The app supports 9 providers with zero setup lock-in.

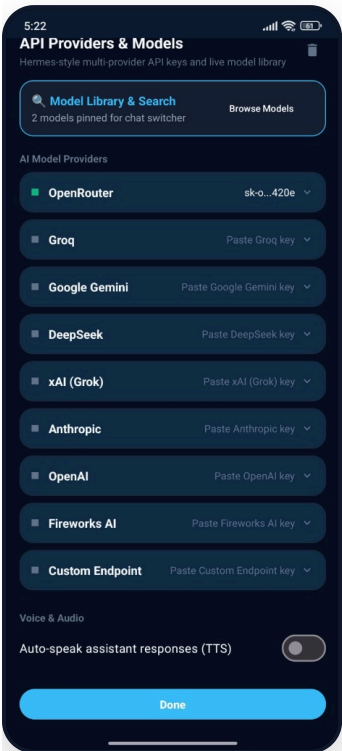


Fig 1.1: API Providers & Models Dialog

Where to Find Free & Supported APIs:

Provider	Website / Portal	Cost & Recommended Models
OpenRouter (Recommended)	<code>openrouter.ai/keys</code>	100% FREE TIER Access to 20+ free models (Nemotron 3, Ling 3.0, Llama 3.3, Qwen 27B).
Google Gemini	<code>aistudio.google.com</code>	FREE GENERATIVE QUOTA Gemini 1.5 Flash (1M token context, high speed).
Groq	<code>console.groq.com/keys</code>	FREE TIER Llama 3.3 70B with ultra-fast 500+ tokens/sec inference.
DeepSeek	<code>platform.deepseek.com</code>	LOW COST PAY-AS-GO DeepSeek-V3 and DeepSeek-R1 reasoning.
OpenAI	<code>platform.openai.com/api-keys</code>	COMMERCIAL API GPT-4o and GPT-4o-mini.
Local / Custom	<code>http://localhost:11434/v1</code>	SELF-HOSTED Local Ollama, vLLM, or LM Studio running on local Wi-Fi.

Step-by-Step Guide: How to Add Your API Key in the App

- 1

Open Settings: Tap the **Gear** (⚙️) icon on the Dashboard or in the Chat top bar.
- 2

Select Your Provider: In the *AI Model Providers* list, tap on your provider (e.g., **OpenRouter** or **Google Gemini**).
- 3

Paste Your Key: Enter or paste your key into the text field (e.g. `sk-or-v1- ...`).
- 4

Verify Green Indicator: A solid **green badge** ■ appears next to the provider name confirming your key is saved.
- 5

(Optional) Enable Voice TTS: Toggle "*Auto-speak assistant responses (TTS)*" to hear responses read aloud.
- 6

Save: Tap **Done** at the bottom. Your AI Engine is immediately **Online & Ready!**

2. Model Library & Search Hub

The app includes an integrated Model Discovery Catalog allowing you to explore over **450+ live models** or quickly pick from **20+ completely free** models.

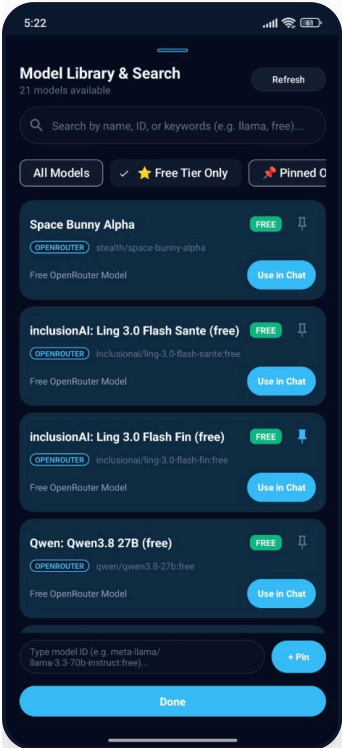


Fig 2.1: Free Tier Only Filter

Navigating the Model Library

To open the Model Hub, tap **Model Library & Search** from the Settings screen or select **Model Library & Hub** from the left navigation drawer.

★ Filter: Free Tier Only

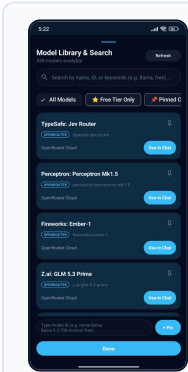
Tap the ★ **Free Tier Only** chip at the top to filter models that require **\$0.00 credits** on OpenRouter:

- **NVIDIA: Nemotron 3 Ultra (free)**: Ultra-capable general reasoning model.
- **inclusionAI: Ling 3.0 Flash Fin / Sante (free)**: Lightweight, lightning-fast response times.
- **Qwen: Qwen3.8 27B (free)**: High-parameter multilingual and mathematical model.
- **Space Bunny Alpha**: Creative synthesis and rapid summarization.

📌 Pinning Models & One-Tap Activation

- **Use in Chat**: Instantly makes the selected model your primary conversational engine and switches back to the chat screen.
- **Pin Icon (📌)**: Pins the model to your *Quick Switcher* bottom sheet so you can toggle between it and other models without re-opening the library.
- **Manual Model Pin**: At the bottom, you can paste any custom model identifier (e.g. `meta-llama/llama-3.3-70b-instruct:free`) and tap + Pin.

🔍 **Live Search**: Type any keyword in the search bar (e.g., "llama", "code", "qwen") to filter the catalog in real-time with sub-millisecond responsiveness.



Full Cloud Directory (458+ Models Available)

Switching to the **All Models** tab provides live access to cutting-edge models from TypeSafe, Fireworks AI, Z.ai GLM, Perceptron, Mistral, and Anthropic. Pinned items are saved in Room DB across sessions.

3. AI Chat Command Center & Quick Model Switcher

The Chat screen combines intelligent generative conversation with seamless multi-model switching, speech-to-text dictation, and speech synthesis.

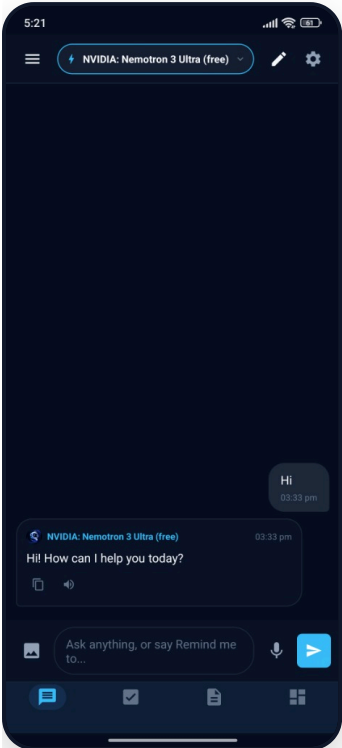
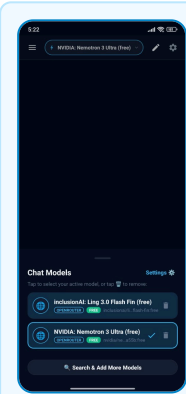


Fig 3.1: Active Chat Screen

Core Chat Interface Elements

- **Active Model Header Chip:** The blue pill at the top displays the active model (e.g., **NVIDIA: Nemotron 3 Ultra (free)**). Tap it to open the quick switcher!
- **Pencil & Gear Icons:** Tap the **Pencil** (✎) to start a clean new chat session; tap the **Gear** (⚙️) to jump directly to API configuration.
- **Message Action Tools:** Every AI response includes two quick action buttons:
 - **Copy Icon** (📄): Copies the complete formatted markdown response to your system clipboard.
 - **Speaker Icon** (🔊): Triggers text-to-speech to read the assistant's answer out loud.
- **Multi-Modal Attachments:** Tap the **Gallery Icon** (🖼️) to attach images for visual analysis.
- **Voice Microphone** (🎤): Tap to dictate your questions or commands hands-free via Android speech recognition.

In-Chat Quick Model Switcher



Frictionless Switching (Fig 3.2)

- Tapping the top header chip opens the *Chat Models* bottom sheet:
- **Tap to Select:** Tap any pinned model to switch immediately. A checkmark (✓) indicates the active model.
 - **Trash Can** (🗑️): Remove a model from your pinned list.
 - **Search & Add:** Opens the full 450+ model catalog to pin new ones.

4. Cyber Dashboard & Morning Audio Briefing

The Dashboard is the dynamic headquarters of your day, providing real-time metric aggregates, audio executive summaries, and schedule tracking.

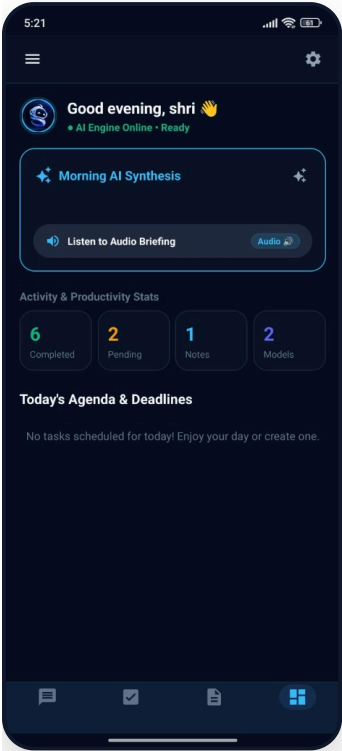


Fig 4.1: Cyber Dashboard Overview

Dashboard Components Breakdown

1. Personalized Greeting & Engine Status

- Displays "Good evening, shri 🖐️", dynamically updating based on system time and authenticated user profile.
- **AI Engine Online • Ready**: Real-time heartbeat checking whether active provider credentials and local databases are operational.

2. Morning AI Synthesis & Audio Briefing

- The **Morning AI Synthesis** card uses LLM reasoning to summarize pending tasks, notes, and calendar deadlines into a 30-second daily briefing.
- Tap "**Listen to Audio Briefing**" 🔊 to listen to your morning schedule read aloud via Android Text-To-Speech while getting ready.

3. Activity & Productivity Stats

Live SQLite counters tracking your organization progress:

6	2	1	2
Completed	Pending	Notes	Models

4. Today's Agenda & Deadlines

Automatically surfaces tasks due before midnight today. If no tasks remain, displays "No tasks scheduled for today! Enjoy your day or create one."

5. AI Workspace & Navigation Drawer

Tapping the top-left hamburger menu (☰) slides open the **AI Workspace Drawer**, serving as your personal account and session manager.

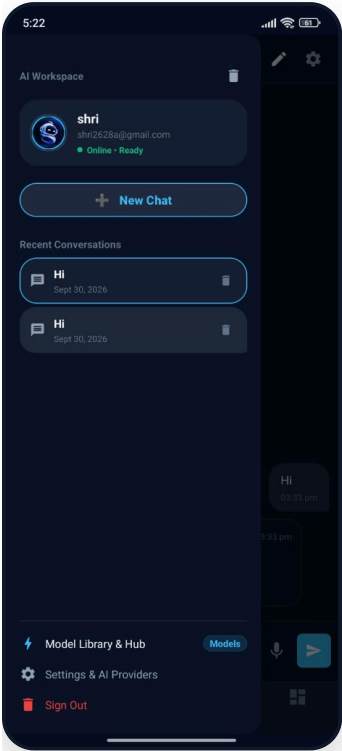


Fig 5.1: AI Workspace Drawer

Drawer Features & Controls

User Profile Card

- Displays custom avatar with neon obsidian cyber ring.
- Shows registered user name (`shri`) and email (`shri2628a@gmail.com`).
- **Online • Ready**: Confirms local database and cloud API connectivity.

+ New Chat Button

Tapping the large pill button initiates an immediate fresh conversation thread, saving the current conversation to history.

Recent Conversations History

- Shows chronological list of past chat sessions (e.g. *"Hi - Sept 30, 2026"*).
- Tap any conversation card to reopen and restore full message history.
- Tap the **Trash Can** () icon on any session to delete that specific transcript from Room SQLite.

Shortcuts & Account Management

-  **Model Library & Hub**: Direct jump to the 450+ model catalog.
-  **Settings & AI Providers**: Fast access to API key management.
-  **Sign Out (Red)**: Securely ends the current session and returns to the authentication screen.

6. Task Matrix & Autonomous Alarms

The **Tasks Matrix** (`TasksFragment.java`) delivers robust schedule planning with priority classification, checkbox states, and system-level alarms.

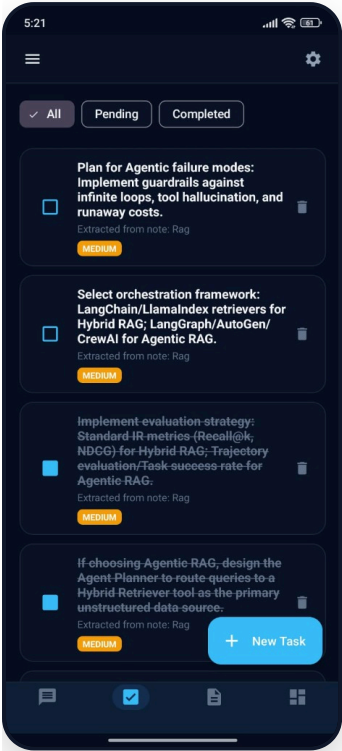


Fig 6.1: Task Matrix with Extracted Tasks

Managing Tasks in the Matrix

1. Filter Tabs: All / Pending / Completed

Quickly toggle between ✓ **All** (shows full list), **Pending** (focuses on unfinished work), and **Completed** (shows strikethrough accomplished items).

2. Priority Badges & Completion Checkboxes

- **Interactive Checkboxes:** Tapping the square checkbox immediately toggles completion status, striking through text and updating dashboard stats.
- **Priority Tags:** Tasks are categorized with color-coded tags: **HIGH** (red), **MEDIUM** (amber), **LOW** (green).

3. AI Smart Task Extraction from Notes!

Notice the subtitle on the tasks in Fig 6.1:
`Extracted from note: Rag`
The assistant allows you to extract actionable todo items from any research note or lecture with a single tap, automatically populating the task matrix!

4. + New Task Floating Action Button

Tap the bright cyan **+ New Task** button at bottom-right to open the creation dialog, set deadlines, and configure exact notification alarms.

7. Knowledge Vault (Notes) & AI Extraction Tools

The **Notes** screen acts as your intellectual repository, enhanced with one-touch AI processing tools for research synthesis.

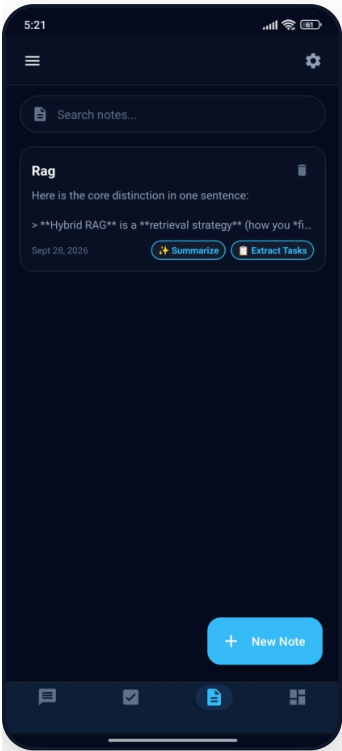


Fig 7.1: Knowledge Vault with AI Tools

Supercharged AI Note Operations

1. One-Tap AI Summarize

Tapping the ✨ **Summarize** chip on any note feeds its content into the active LLM engine to extract key executive takeaways in 3 concise bullet points.

2. One-Tap Extract Tasks

Tapping 📄 **Extract Tasks** automatically parses the note's action items, architectural plans, and deliverables, and converts them directly into records in your **Task Matrix**!

3. Live Full-Text Search

Use the top **Search notes...** bar to instantly filter across note titles, body text, and creation dates.

+ 4. Rich Note Creation (+ New Note)

Tap the floating cyan + **New Note** button to compose research documentation, lecture notes, or code snippets with markdown preview.

5. Clean Management & Deletion

Tap the trash icon on the top-right of any note card to remove it permanently from local Room database storage.

8. Natural Language Voice Commands Cheat Sheet

Tap the **Microphone** (🎤) on any screen and speak these natural phrases to execute actions instantly:

Target Action	Voice / Natural Language Command	Resulting App Behavior
Create Urgent Task	"Add task submit lab journal tomorrow 10am urgent"	Creates HIGH priority task, sets due date, registers exact alarm.
Set Reminder	"Remind me to attend seminar tonight at 8 PM"	Parses "tonight 8 PM", sets exact alarm, confirms in chat.
General Task	"New task buy circuit components for IoT project"	Creates standard task with MEDIUM priority in Task Matrix.
Quick Note	"Take a note: Hybrid RAG is a retrieval strategy"	Saves note directly to Knowledge Vault with timestamp.
Complex Query	"Compare Agentic RAG vs Standard RAG architecture"	Routes query to active cloud LLM for full technical synthesis.

Troubleshooting & Permissions Guide

 **Alarm / Notification Not Ringing?**

1. Open Android Settings → Apps → **AI Personal Assistant**.

2. Set Battery Usage to "**Unrestricted**" so Android Doze Mode doesn't pause alarms.

3. Ensure "**Allow notifications**" is toggled ON.

 **Speech Recognition Error?**

1. Go to App Permissions → **Microphone** → Select "*Allow only while using the app*".

2. Verify Google Speech Services / Voice typing is enabled in Android Keyboard settings.

 **Switching Back to Offline Mode:** If you run out of internet or API credits, the local deterministic engine continues to parse all tasks, notes, and reminders automatically without showing errors.

AI Personal Assistant v1.0 • Academic Project Build

com.mitaoe.shridhar202401040197